ANSYS PathFinder ESD Integrity Analysis

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1 Purpose of This Document

The goal of this document is to enable a user to achieve functional safety compliance satisfying ISO26262 of TCL1 (Tool Confidence Level 1) in Power Integrity and Reliability, using software product ANSYS PathFinder™. This document is a report classifying used tools according to their confidence need during the use of a software tool chain or parts of a software tool chain. The document consists of use-cases, and error scenarios which may happen during the analysis with PathFinder and measures for detecting the tool errors/malfunctions through FMEA (Failure Models and Effects Analysis). We recommend that the user read respective manuals of the product for additional information beyond this safety manual.

PathFinder™:

“Chapter 16: Pathfinder™ ESD Analysis” in RedHawk_UserMan_19.1.pdf (or)

“Chapter 12: Pathfinder™ ESD Analysis” in Totem_UserMan_19.1.pdf

2 Terms and Definitions

All abbreviations used in the document and full name of each abbreviation.

Abbreviated Terms	Full Name
ESD	Electro Static Discharge
CD	Current Density
EM	Electromigration
GDSII	Graphic Data System II
DEF	Design Exchange Format
LEF	Library Exchange Format
LVS	Layout Vs Schematic
PDN	Power Delivery Network
PLOC	Pad Location Information
P2P	Point-to-Point
DC	Direct Current
IV curve	Current-Voltage curve

3 ANSYS PathFinder- ESD Integrity Analysis

3.1 Introduction

ANSYS's PathFinder™ is a ESD integrity analysis tool which is an option to RedHawk and Totem platforms, as shown in the Figure 1.

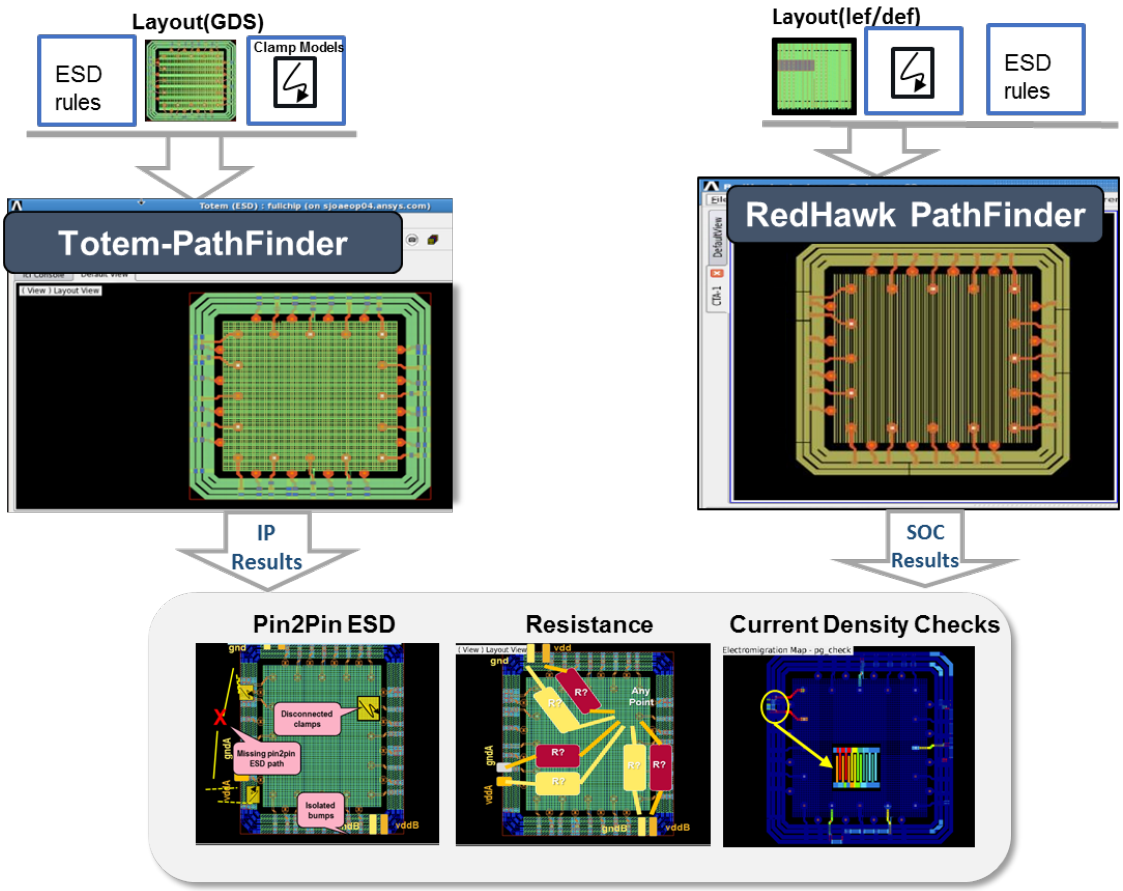

Figure 1 PathFinder ESD Analysis in RedHawk and Totem

RedHawk platform is used to analyze power noise and reliability analysis on System on chip and Totem performs the same on Analog and Mixed Signal designs. This document only covers the ESD analysis capability available in RedHawk and Totem platform with PathFinder option.

3.2 Product Version

PathFinder product is an add-on module (*aka* option) and is available as a part of RedHawk or Totem installation. RedHawk and Totem 19.1 and subsequent versions are in production use at the time of certification. This document remains valid for upcoming tool versions if the product changes do not impact the listed use cases, error scenarios, and error detection and prevention methods.

3.3 PathFinder- Design Input Data Requirements

PathFinder uses industry-standard design input data for analyzing ESD integrity on SoC, Analog and mixed signal designs. The design data needed are described in the following table.

File	Description
Layout	Physical description of the design in different formats (**.gds, OAS, CCI*) Digital blocks can be represented as LEF/DEF database
Netlist	Simulation Netlist in spice/spectre formats along with parasitic extracted netlist in industry standard format like DSPF
Clamp info file	Description of ESD devices containing the anode/cathode and pin-pairs and ON resistance or IV curves
Power/Ground and signal pad nets	Power/Ground net names, supply voltage, etc
Technology file (**.tech*)	Interconnect parasitic properties (RC) and dielectric information (obtained from Foundry)
Pad files (*.ploc)	Power/Ground and Signal pad X, Y location and layers
ESD rule	Rule containing ESD zap combinations that needs to be tested

3.4 PathFinder- Work Flow

The Overall work-flow involves the following steps.

(a) Design import:

This includes translation of layout data and recognize power/ground and signal interconnect, Clamp devices and attaching corresponding ON resistance or IV curve models.

(b) Parasitic Extraction:

This step includes Power/Ground and signal rail parasitic resistance extraction for further analysis.

(c) Connectivity Checks:
This step includes checking if there is a ESD discharge path through ESD clamps between any set of pads in the design.

(d) Rule based checks:

This step includes performing P2P resistance checking and DC simulation to determine current flowing through metal interconnects and perform current density checks on the same.

(e) Violation Reporting

This step includes reporting P2P resistance violations and current density violations and other design weakness like disconnects or high resistance paths in the design.

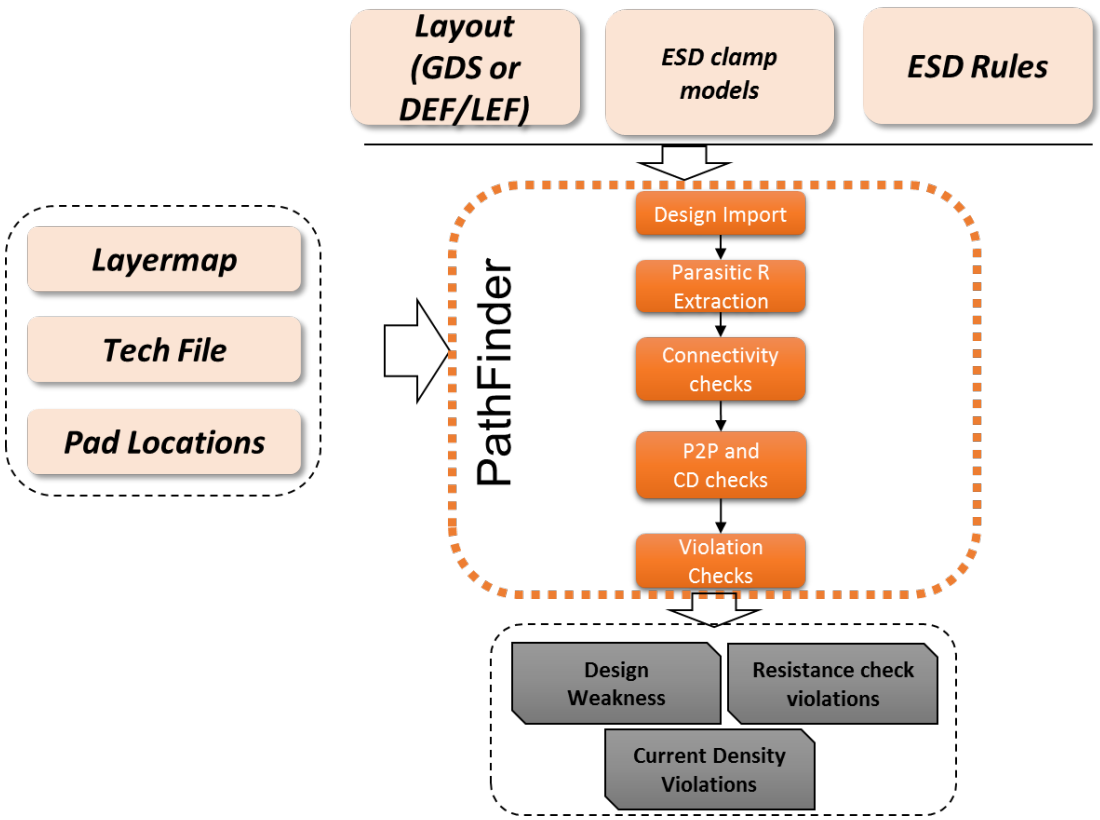

Figure 2 PathFinder work flow

4 Use Case (UC1): ESD Connectivity Checks

Using PathFinder, the user can verify the basic grid quality. In this analysis, the user can find:

1. Unconnected ESD devices from signal, power and ground nets
2. Pads that are not connected to any ESD devices

This analysis can be done with minimal set of data. In this analysis, PathFinder performs the power and ground network extraction, ESD device(clamp) recognition based on user defined electrical information and physical cell names. It identifies the ESD devices that are not physically connected to the power/ground network in this process or any power/ground or signal pads that are not connected to any ESD devices. PathFinder also performs an exhaustive check of ESD discharge path between any 2 set of pads in the design and report any set of pads that do not have a ESD discharge pathways through the clamp. PathFinder can detect direct or indirect connections (through many stages or ESD devices (aka stacked ESD devices) between two pads.

To do this analysis you need to specify all the required inputs to the tool to setup the design and specify the analysis condition in a configuration file and execute the flow. PathFinder provides violations as a report with the co-ordinates of violations for users to fix the design weaknesses. Here is the flow diagram describing the work-flow for Grid Weakness Analysis

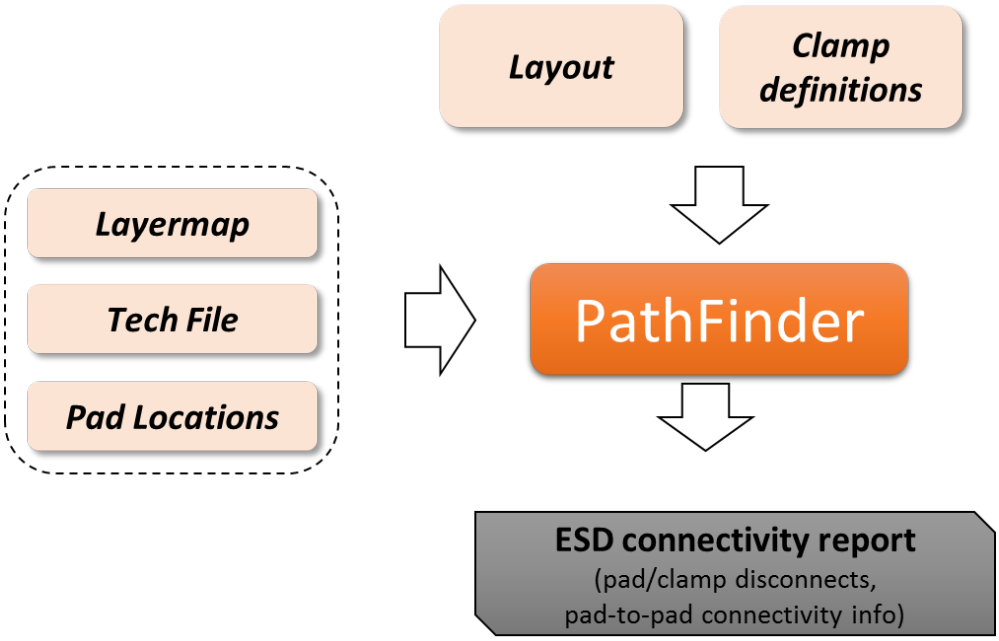

Figure 3 Work flow for ESD Connectivity Checks

5 Use Case (UC2): P2P Resistance Checks

Using PathFinder, the user can perform P2P resistance checks. In this analysis, the user can find:

1. Effective resistance between power/ground/signal pads and the ESD protection devices
2. Effective resistance between each pair of Power/ground and signal pads through ESD protection devices
3. Check against user specified thresholds and report any violations

In this analysis, the tool requires on layout data and technology collaterals, power/ground signal pads, ESD device definitions as clamp info and ESD rules w/ type of resistance checks and maximum resistance constraints. The scope of this document restricted to provide overview of the work-flow and does not cover the details of ESD rules and clamp definitions. Users are advised to refer to “Pathfinder™ ESD Analysis” chapter in RedHawk or Totem User Manual and consult with ANSYS Tool Applications Engineer for further advice.

PathFinder performs the power and ground network extraction, ESD device recognition based on technology collaterals and design data. It assigns the IV curve or ON resistance model for ESD devices based on the definitions provided by the user. PathFinder uses type of ESD rules and constraints provided in ESD rules to check for resistance between pad and ESD devices or between different set of pads. To do this analysis you need to specify all the required inputs to the tool to setup the design and specify the respective ESD rules in a configuration file and execute the flow. PathFinder provides violations as color coded maps overlaid on layout or as a report with the paths that are violating the user specified maximum resistance thresholds.

Here is the flow diagram describing the work-flow for P2P Resistance checks.

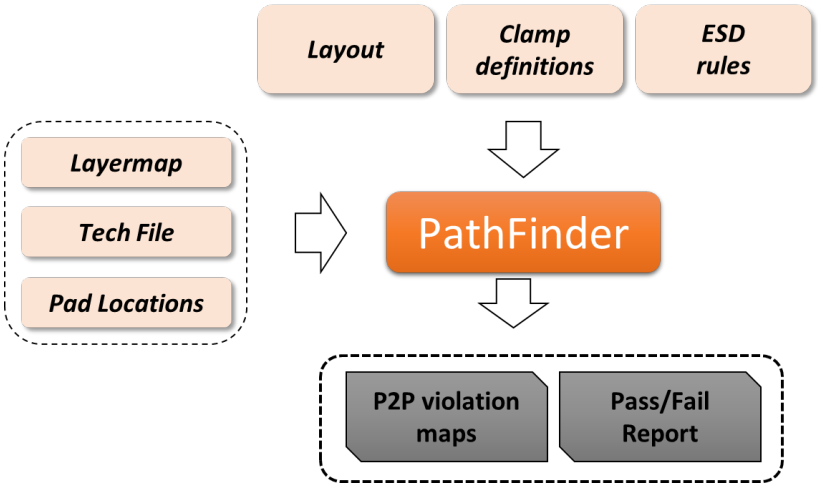

Figure 4 Work flow for P2P Resistance checks

6 Use Case (UC3): ESD Current Density checks

Using PathFinder, the user can Current density analysis. In this analysis, the user can find:

1. Narrow metal segments and insufficient vias in ESD discharge paths
2. High voltage drops on ESD discharge path that can cause ESD issues

In this analysis, the tool requires on layout data and technology collaterals, power/ground signal pads, ESD device definitions as clamp info and ESD rules w/ type of current density checks performed and value of the current injected on pad or ESD device. The technology collateral should also contain ESD current density limits for the metal and via layers for performing this check. The scope of this document restricted to provide overview of the work-flow and does not cover the details of ESD rules and clamp definitions. Users are advised to refer to "Pathfinder™ ESD Analysis" chapter in RedHawk or Totem User Manual and consult with ANSYS Tool Applications Engineer for further advice.

PathFinder performs the power and ground network extraction, ESD device recognition based on technology collaterals and design data. It assigns the IV curve or ON resistance model for ESD devices based on the definitions provided by the user. PathFinder uses type of ESD rules provided in ESD rules to inject user specified current between pad and ESD devices or between different set of pads. PathFinder also checks every geometry in the ESD discharge path against maximum current limit in the technology file to determine if the interconnect can carry the current without any melt-down during ESD discharge. To do this analysis you need to specify all the required inputs to the tool to setup the design and specify the respective ESD rules in a configuration file and execute the flow. PathFinder provides violations as color coded maps overlaid on layout or as a report with the paths that are violating the current density limits in the technology file. Here is the flow diagram describing the work-flow for ESD current density checks.

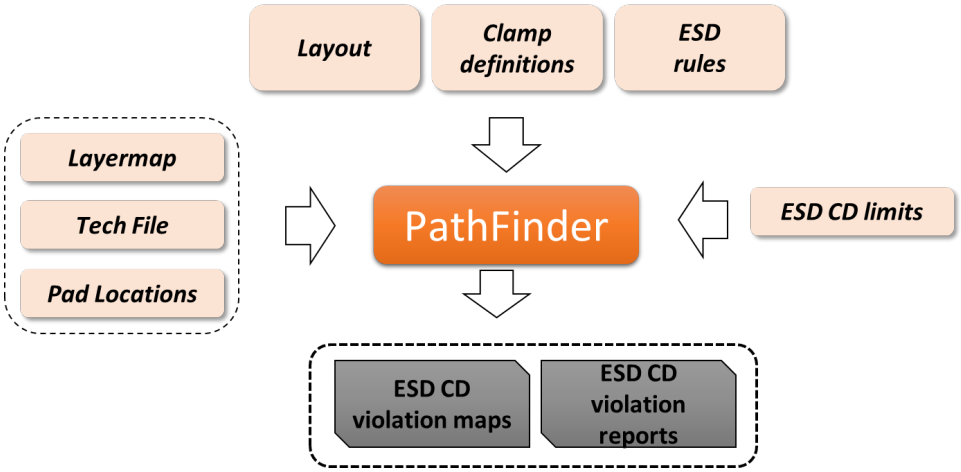

Figure 5 ESD Current density checks

7 Error Scenario

7.1 Failure Mode and Effect Analysis

Use Case	Error Condition	Potential Malfunctions and Erroneous reporting	Tool Impact (TI)	**Justification for TI**	Tool Error Detection (TD)	Justification for TD	Error Detection and Prevention Methods	Tool Classification Level (TCL)
UC1	EC1	Tool does not identify ESD devices that are not connected to the power/ground or signal nets or pads not connected to ESD devices	TI2	Missed design weakness	TD1	Error can be detected by additional checks	Please refer to section 8.1.1 for details	TCL1
UC2	EC2	Unreasonably large resistance values due to inputs, constraints or tool issues.	TI2	Wrong analysis results in tool reports can lead to undetected ESD issues	TD1	Error can be detected by additional checks	Please refer to section 8.2.1 for details	TCL1
UC2	EC3	Very low or no violations detected by the tool	TI2	Wrong analysis results in tool reports can lead to undetected ESD issues	TD1	Error can be detected by additional checks	Please refer to section 8.2.2 for details	TCL1
UC3	EC4	Unreasonably large Current density violations due to inputs, constraints or tool issues.	TI2	Wrong analysis results in tool reports can lead to undetected ESD issues	TD1	Error can be detected by additional checks	Please refer to section 8.3.1 for details	TCL1
UC3	EC5	Very low or no current density violations reported by the tool	TI2	Wrong analysis results in tool reports can lead to undetected ESD issues	TD1	Error can be detected by additional checks	Please refer to section 8.3.2 for details	TCL1

Table 1 Failure Mode and Effect Analysis

8 Error Detection and Prevention Methods

8.1 Use Case (UC1)– ESD connectivity checks

8.1.1 Error Condition (EC1)

To avoid failures caused by undetected design weakness, either due to PathFinder tool deficiencies, incorrect user inputs or constraints it is recommended to perform the following detection and prevention measures.

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis
2. Any disconnects in the design should be caught through LVS.
3. Perform resistance and current density checks. Design weakness may manifest as paths with high resistance or paths with higher current density violations.
4. Perform ESD checks for HBM/CDM of the chip and chip in package for intended target ESD levels and functional validation after ESD zaps to ensure there are no ESD issues. ESD failures manifest as functional failure, high power consumption due to high leakage or timing/performance issues after ESD testing.

8.2 Use Case (UC2) – P2P Resistance Checks

8.2.1 Error Condition (EC2)

To avoid failures caused by undetected design weakness, either due to PathFinder tool deficiencies, incorrect user inputs or constraints it is recommended to perform the following detection and prevention measures.

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis
2. Add margin and additional ESD protection devices as necessary
3. Perform current density checks. Design weakness may manifest as paths with high voltage drop or paths with higher current density violations.
4. Perform ESD checks for HBM/CDM of the chip and chip in package for intended target ESD levels and functional validation after ESD zaps to ensure there are no ESD issues. ESD failures manifest as functional failure, high power consumption due to high leakage or timing/performance issues after ESD testing.

8.2.2 Error Condition (EC3)

To avoid failures caused by undetected design weakness, either due to PathFinder tool deficiencies, incorrect user inputs or constraints it is recommended to perform the following detection and prevention measures.

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis
2. Add margin and additional ESD protection devices as necessary
3. Perform current density checks. Design weakness may manifest as paths with high voltage drop or paths with higher current density violations.
4. Perform ESD checks for HBM/CDM of the chip and chip in package for intended target ESD levels and functional validation after ESD zaps to ensure there are no ESD issues. ESD failures manifest as functional failure, high power consumption due to high leakage or timing/performance issues after ESD testing.

8.3 Use Case (UC3) – ESD Current density checks

8.3.1 Error Condition (EC4)

To avoid failures caused by undetected design weakness, either due to PathFinder tool deficiencies, incorrect user inputs or constraints it is recommended to perform the following detection and prevention measures.

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis
2. Add margin and additional ESD protection devices as necessary
3. Perform current density checks. Design weakness may manifest as paths with high voltage drop or paths with higher current density violations.
4. Perform ESD checks for HBM/CDM of the chip and chip in package for intended target ESD levels and functional validation after ESD zaps to ensure there are no ESD issues. ESD failures manifest as functional failure, high power consumption due to high leakage or timing/performance issues after ESD testing.

8.3.2 Error Condition (EC5)

To avoid failures caused by undetected design weakness, either due to PathFinder tool deficiencies or incorrect user inputs and/or constraints it is recommended to perform the following detection and prevention measures.

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis
2. Add margin and additional ESD protection devices as necessary
3. Perform current density checks. Design weakness may manifest as paths with high voltage drop or paths with higher current density violations.
4. Perform ESD checks for HBM/CDM of the chip and chip in package for intended target ESD levels and functional validation after ESD zaps to ensure there are no ESD issues. ESD failures manifest as functional failure, high power consumption due to high leakage or timing/performance issues after ESD testing.